HEISENBERG UNCERTAINTY AND TWISTED MÖBIUS DYNAMICS IV: COHOMOLOGICAL CURVATURE, PRIME–FACTOR VARIANCE, AND A ZETA–ZERO SINGULARITY CRITERION

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Remark 0.1 (Version 4). The canonical pair, Weyl calculus, Robertson inequality, and general twisted algebra are standard. The arithmetic specialization $n \mapsto (\ell \log n, \kappa\Omega(n))$ produces the

specific cocycle studied here. Version 4 adds three exact results: a structural obstruction showing that every expectation value of a linearly Möbius-weighted state is necessarily quadratic in the Möbius coefficients; a closed formula for the second ϑ-curvature of the twisted inverse $z_\vartheta(n)$; and an exact Dirichlet-series summation of that curvature which continues locally to $\Re s > 1/2$ and develops a logarithmic branch singularity at every Riemann zero in that half-plane.

1. Introduction: the Version 4 structure

Version 1 placed a standard Heisenberg pair next to a zeta multiplier but did not couple the two structures. Version 2 corrected that defect: the exact canonical commutation relation deformed the arithmetic product itself through a nontrivial cocycle. The Version 2 audit nevertheless identified two remaining weaknesses. First, the actual uncertainty principle was not stated prominently. Second, several domain, inverse, and convergence claims required repair.

Version 4 starts from the self-adjoint position and momentum operators

$$(Q\psi)(q) = q\psi(q), \qquad (P\psi)(q) = -i\hbar\psi'(q), \tag{1.1}$$

with the canonical commutation relation

$$[Q, P] = i\hbar I \tag{1.2}$$

on the Schwartz core. For every normalized Schwartz state, the Heisenberg uncertainty principle is

$$\boxed{\Delta_\psi Q \, \Delta_\psi P \geq \frac{\hbar}{2}.} \tag{1.3}$$

Here the uncertainty is a mathematical spread intrinsic to the state, not an experimental error.

The arithmetic specialization assigns to an integer n the phase-space increments

$$x_n = \ell \log n, \qquad y_n = \kappa\Omega(n), \tag{1.4}$$

and the self-adjoint generator

$$G_n = y_n Q - x_n P. \tag{1.5}$$

The first central statement of Version 4 is an uncertainty law for two integers:

$$\boxed{\Delta_\psi G_m \, \Delta_\psi G_n \geq \frac{\hbar\ell\kappa}{2} \left|\Omega(m)\log n - \log m\, \Omega(n)\right|.} \tag{1.6}$$

For distinct primes p, q, this becomes

$$\boxed{\Delta_\psi G_p \, \Delta_\psi G_q \geq \frac{\hbar\ell\kappa}{2} \left|\log \frac{q}{p}\right|.} \tag{1.7}$$

Thus the same arithmetic symplectic area that later appears in the twisted Möbius product already appears as a lower bound forced directly by the Heisenberg uncertainty principle.

The associated Weyl operators are

$$W_n = \exp\left(\frac{i}{\hbar} G_n\right), \tag{1.8}$$

and satisfy

$$W_m W_n = \chi_\vartheta(m, n) W_{mn}, \tag{1.9}$$

where

$$\chi_\vartheta(m, n) = \exp\left[\frac{i\vartheta}{2}\big(\Omega(m)\log n - \log m\, \Omega(n)\big)\right], \qquad \vartheta = \frac{\ell\kappa}{\hbar}. \tag{1.10}$$

The cocycle is proved below to be cohomologically nontrivial for every $\vartheta \neq 0$. It induces an associative noncommutative Dirichlet product. Möbius inversion in that product defines coefficients $z_\vartheta(n)$. For prime powers and distinct prime pairs one obtains

$$z_\vartheta(p^r) = 1, \tag{1.11}$$

$$z_\vartheta(pq) = 2\cos\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) - 1. \tag{1.12}$$

The same determinant-like arithmetic quantity therefore controls both the uncertainty lower bound (1.6) and the phase responsible for the deformed inverse.

Version 4 also differentiates the twisted inverse itself at the commuting point. For

$$n = \prod_{j=1}^{r} p_j^{a_j}, \qquad T(a) = \frac{a(a+1)}{2}, \tag{1.13}$$

one obtains

$$\boxed{-z_0''(n) = \frac{1}{2}\sum_{1\le i<j\le r} T(a_i)T(a_j)\log^2\frac{p_j}{p_i}.} \tag{1.14}$$

For square-free $n = p_1\cdots p_r$,

$$\boxed{-z_0''(n) = \frac{1}{2}\left[r\sum_{p\mid n}(\log p)^2 - (\log n)^2\right] = \frac{r^2}{2}\operatorname{Var}_{p\mid n}(\log p).} \tag{1.15}$$

Summing this curvature does not return a two-variable Möbius correlation. For $\Re s > 1$,

$$\mathcal{K}(s) = -\sum_{n\ge 1}\frac{z_0''(n)}{n^s} = \frac{\zeta(s)}{2}\big(S_0(s)S_2(s) - S_1(s)^2\big), \tag{1.16}$$

where

$$S_j(s) = \sum_p \frac{p^{-s}}{(1-p^{-s})^2}(\log p)^j. \tag{1.17}$$

Section 10 rewrites these functions through $\log\zeta(rs)$ and $\zeta'(rs)/\zeta(rs)$, producing a local continuation to $\Re s > 1/2$ and an exact singularity criterion for nontrivial zeta zeros to the right of the critical line.

Finally, a bounded self-adjoint Möbius Hamiltonian is defined for $\sigma > 1$. Its Schrödinger evolution and the exact Heisenberg equations for $Q(t)$ and $P(t)$ are derived with domains stated explicitly. The paper does not claim a connection to RH; analytic continuation, functional equations, and spectral information remain open.

Remark 1.1 (Scope of Version 4). The operator deformation is developed in the half-planes stated for each operator. Version 4 additionally derives a direct local relation between the second deformation curvature and the classical zero set in Section 11.

2. Exact canonical kinematics

2.1. Hilbert space, position, and momentum.

Definition 2.1 (Hilbert space)**.** Fix $\hbar > 0$ and set

$$\mathcal{H} = L^2(\mathbb{R}, dq) \tag{2.1}$$

with its standard complex Hilbert-space inner product.

Definition 2.2 (Position)**.** The position operator Q is

$$(Q\psi)(q) = q\psi(q) \tag{2.2}$$

with maximal domain

$$\mathcal{D}(Q) = \{\psi \in L^2(\mathbb{R}) : q\psi(q) \in L^2(\mathbb{R})\}. \tag{2.3}$$

Definition 2.3 (Momentum)**.** The momentum operator P is

$$(P\psi)(q) = -i\hbar\,\psi'(q) \tag{2.4}$$

with domain

$$\mathcal{D}(P) = H^1(\mathbb{R}). \tag{2.5}$$

Proposition 2.4 (Self-adjointness and common core)**.** *The operators Q and P are self-adjoint. The Schwartz space $\mathcal{S}(\mathbb{R})$ is a common invariant core for both.*

Proof. Q is the maximal multiplication operator by the real function q. Under the unitary Fourier transform with convention

$$(\mathcal{F}\psi)(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} e^{-ipq/\hbar}\psi(q)\,dq, \tag{2.6}$$

P is unitarily equivalent to multiplication by the real variable p. The core statement follows by standard cutoff and mollification arguments. See, for example, [3]. □

Proposition 2.5 (Exact canonical commutation relation)**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$,*

$$[Q, P]\psi = i\hbar\psi. \tag{2.7}$$

Proof. Direct differentiation gives

$$(QP\psi)(q) = -i\hbar q\psi'(q), \tag{2.8}$$

$$(PQ\psi)(q) = -i\hbar\psi(q) - i\hbar q\psi'(q). \tag{2.9}$$

Subtracting (2.9) from (2.8) proves (2.7). □

2.2. **The Heisenberg uncertainty principle and exact incompatibility.**

Definition 2.6 (Expectation and uncertainty)**.** Let A be self-adjoint and let $\psi \in \mathcal{D}(A)$ be normalized. Define

$$\langle A\rangle_\psi = \langle \psi, A\psi\rangle, \tag{2.10}$$

$$(\Delta_\psi A)^2 = \|(A - \langle A\rangle_\psi)\psi\|^2. \tag{2.11}$$

Theorem 2.7 (Heisenberg uncertainty principle)**.** *For every normalized $\psi \in \mathcal{S}(\mathbb{R})$,*

$$\boxed{\Delta_\psi Q\,\Delta_\psi P \geq \frac{\hbar}{2}.} \tag{2.12}$$

Proof. Set

$$\widetilde{Q} = Q - \langle Q\rangle_\psi I, \qquad \widetilde{P} = P - \langle P\rangle_\psi I. \tag{2.13}$$

By Cauchy–Schwarz,

$$\Delta_\psi Q\,\Delta_\psi P \geq \left|\langle \widetilde{Q}\psi, \widetilde{P}\psi\rangle\right|. \tag{2.14}$$

The modulus dominates the absolute value of the imaginary part, and on $\mathcal{S}(\mathbb{R})$

$$2i\ \mathrm{Im}\langle \widetilde{Q}\psi, \widetilde{P}\psi\rangle = \langle \psi, [Q, P]\psi\rangle \tag{2.15}$$

$$= i\hbar. \tag{2.16}$$

Combining (2.14)–(2.16) proves (2.12). □

Remark 2.8 (Meaning of the inequality). Equation (2.12) is the Heisenberg uncertainty principle used in this paper. The quantities $\Delta_\psi Q$ and $\Delta_\psi P$ are intrinsic standard deviations of a state. They are not measurement errors. Exact simultaneous localization would require both to vanish, which is prohibited by (2.12).

Theorem 2.9 (No simultaneous exact position and momentum)**.** *There is no nonzero state with simultaneous exact position and exact momentum. More precisely, both of the following are impossible for real q_0, p_0:*

(1) a nonzero vector ψ in a common invariant domain satisfying

$$Q\psi = q_0\psi, \qquad P\psi = p_0\psi; \tag{2.17}$$

(2) a nonzero tempered distribution $T \in \mathcal{S}'(\mathbb{R})$ satisfying

$$qT = q_0 T, \qquad -i\hbar T' = p_0 T. \tag{2.18}$$

Proof. For vectors, (2.17) gives $[Q,P]\psi = 0$, whereas the CCR gives $[Q,P]\psi = i\hbar\psi$; hence $\psi = 0$. For distributions define directly

$$QT = qT, \qquad PT = -i\hbar T'. \tag{2.19}$$

The distributional Leibniz rule gives

$$[Q,P]T = i\hbar T. \tag{2.20}$$

The simultaneous eigenvalue equations would make the left-hand side zero, so $T = 0$. □

3. Weyl displacements and the arithmetic phase space

3.1. **Weyl operators.**

Definition 3.1 (Weyl displacement)**.** For $(x,y) \in \mathbb{R}^2$, define

$$W(x,y) = \exp\left(\frac{i}{\hbar}(yQ - xP)\right). \tag{3.1}$$

This is a unitary operator on all of $\mathcal{H}$.

Proposition 3.2 (Coordinate action)**.** *For $\psi \in L^2(\mathbb{R})$,*

$$(W(x,y)\psi)(q) = \exp\left(\frac{i}{\hbar} y\left(q - \frac{x}{2}\right)\right)\psi(q-x) \tag{3.2}$$

almost everywhere.

Proof. Since $[iyQ/\hbar, -ixP/\hbar]$ is scalar, the Baker–Campbell–Hausdorff formula gives

$$W(x,y) = e^{-ixy/(2\hbar)} e^{iyQ/\hbar} e^{-ixP/\hbar}. \tag{3.3}$$

The last factor translates by x and the middle factor multiplies by $e^{iyq/\hbar}$, yielding (3.2). □

Proposition 3.3 (Weyl form of the CCR)**.** *Let $z = (x,y)$ and $z' = (x',y')$. Define*

$$\omega(z,z') = yx' - xy'. \tag{3.4}$$

Then

$$W(z)W(z') = \exp\left(\frac{i}{2\hbar}\omega(z,z')\right) W(z+z'). \tag{3.5}$$

Consequently,

$$[W(z), W(z')] = 2i \sin\left(\frac{\omega(z,z')}{2\hbar}\right) W(z+z'). \tag{3.6}$$

Proof. For

$$A_z = \frac{i}{\hbar}(yQ - xP), \tag{3.7}$$

the CCR gives

$$[A_z, A_{z'}] = \frac{i}{\hbar}\omega(z, z')I. \tag{3.8}$$

This commutator is central, so the Baker–Campbell–Hausdorff series stops after the commutator term and gives (3.5). Reversing z and z' changes the sign of ω, giving (3.6). The Weyl relations are standard; see [2]. □

Proposition 3.4 (Position and momentum displacement equations)**.** *On* $\mathcal{S}(\mathbb{R})$*,*

$$[Q, W(x, y)] = xW(x, y), \tag{3.9}$$

$$[P, W(x, y)] = yW(x, y). \tag{3.10}$$

Proof. Both identities follow by differentiating (3.2), or directly from (2.7) because the relevant commutators with the exponent are scalar. □

3.2. **Arithmetic embedding.**

Definition 3.5 (The additive prime-factor counter)**.** For

$$n = \prod_p p^{v_p(n)}, \tag{3.11}$$

define

$$\Omega(n) = \sum_p v_p(n). \tag{3.12}$$

Then

$$\Omega(mn) = \Omega(m) + \Omega(n). \tag{3.13}$$

Definition 3.6 (Arithmetic phase-space map)**.** Fix two positive scales $\ell > 0$ and $\kappa > 0$. For $n \in \mathbb{N}$, put

$$x_n = \ell \log n, \tag{3.14}$$

$$y_n = \kappa\Omega(n), \tag{3.15}$$

$$z_n = (x_n, y_n). \tag{3.16}$$

Define the dimensionless Heisenberg area parameter

$$\vartheta = \frac{\ell\kappa}{\hbar}. \tag{3.17}$$

Remark 3.7 (Why these two coordinates). The first coordinate is forced by multiplicative size because $\log(mn) = \log m + \log n$. The second coordinate is chosen as the simplest integer-valued completely additive function that directly controls the Möbius sign on square-free integers:

$$\mu(n) \neq 0 \quad \Longrightarrow \quad \mu(n) = (-1)^{\Omega(n)}. \tag{3.18}$$

The construction can be generalized by replacing Ω with another completely additive arithmetic function, but the present notebook fixes Ω throughout.

Definition 3.8 (Arithmetic Weyl operators)**.** For $n \in \mathbb{N}$, define

$$W_n = W(x_n, y_n). \tag{3.19}$$

Theorem 3.9 (Exact Heisenberg cocycle on the integers)**.** *For $m, n \in \mathbb{N}$,*

$$W_m W_n = \chi_\vartheta(m,n) W_{mn}, \tag{3.20}$$

where

$$\chi_\vartheta(m,n) = \exp\left[\frac{i\vartheta}{2}\big(\Omega(m)\log n - \log m\,\Omega(n)\big)\right]. \tag{3.21}$$

Moreover,

$$[W_m, W_n] = 2i\sin\left[\frac{\vartheta}{2}\big(\Omega(m)\log n - \log m\,\Omega(n)\big)\right] W_{mn}. \tag{3.22}$$

Proof. Complete additivity gives

$$z_{mn} = z_m + z_n. \tag{3.23}$$

Substitution of (3.14)–(3.17) into (3.5) gives (3.20); (3.22) follows from (3.6). □

Corollary 3.10 (Prime commutator)**.** *For distinct primes p, q,*

$$[W_p, W_q] = 2i\sin\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) W_{pq}. \tag{3.24}$$

Thus the prime operators commute only at the resonant values

$$\vartheta \log\frac{q}{p} \in 2\pi\mathbb{Z}. \tag{3.25}$$

Remark 3.11 (Where Heisenberg enters nontrivially)*.* Equation (3.20) is the first point where Version 2 ceases to be ordinary Dirichlet algebra in operator notation. If the canonical pair were commuting, the Weyl cocycle would be 1 and the arithmetic product below would reduce to ordinary Dirichlet convolution. For nonzero ϑ, the CCR creates order-dependent phases between different arithmetic factors.

Theorem 3.12 (The arithmetic cocycle is not gauge-trivial)**.** *Extend* log *and* Ω *additively to* $\mathbb{Q}_{>0}^\times$*. For every real $\vartheta \neq 0$, the cocycle χ_ϑ is not a scalar coboundary on $\mathbb{N}^\times$, and its extension is not a scalar coboundary on $\mathbb{Q}_{>0}^\times$.*

Proof. On a commutative semigroup, every scalar coboundary

$$(\delta a)(m,n) = a(m)a(n)a(mn)^{-1} \tag{3.26}$$

is symmetric in m, n. Hence a necessary condition for χ_ϑ to be a coboundary is

$$\frac{\chi_\vartheta(m,n)}{\chi_\vartheta(n,m)} = 1 \tag{3.27}$$

for all m, n. For distinct primes,

$$\frac{\chi_\vartheta(p,q)}{\chi_\vartheta(q,p)} = \exp\left(i\vartheta\log\frac{q}{p}\right). \tag{3.28}$$

If this were one simultaneously for $(p,q) = (2,3)$ and $(2,5)$, then there would be nonzero integers k, l such that

$$\vartheta\log(3/2) = 2\pi k, \qquad \vartheta\log(5/2) = 2\pi l. \tag{3.29}$$

Thus $\log(3/2)/\log(5/2) = k/l \in \mathbb{Q}$. Exponentiating a reduced rational relation would produce an equality between incompatible prime factorizations, contradicting unique factorization. Therefore (3.27) cannot hold for all primes. □

4. Structural filter: linear Möbius states cannot escape second order

Let $v_n \in \mathcal{H}$ be arbitrary vectors and define

$$\Psi_x = \sum_{n \le x} \mu(n) v_n. \tag{4.1}$$

Theorem 4.1 (Quadratic obstruction for all expectation-value routes)**.** *For every linear operator A for which the expression is defined,*

$$\langle \Psi_x, A\Psi_x \rangle = \sum_{m,n \le x} \mu(m)\mu(n) \langle v_m, A v_n \rangle. \tag{4.2}$$

Hence every norm, expectation value, covariance, variance, or Robertson bound obtained from a state constructed linearly with coefficients $\mu(n)$ is a quadratic Möbius form.

Proof. Expand the inner product sesquilinearly and use that $\mu(n)$ is real. □

Corollary 4.2 (State-superposition route closed)**.** *Changing the reference vector, the vectors v_n, or the observable A does not alter the quadratic dependence in* (4.2)*. A non-quadratic consequence must therefore be extracted from the algebraic deformation itself.*

5. Arithmetic uncertainty generated by the CCR

Definition 5.1 (Integer generator)**.** For $n \ge 1$, define on $\mathcal{S}(\mathbb{R})$ the symmetric operator

$$G_n^{(0)} = y_n Q - x_n P = \kappa \Omega(n) Q - \ell \log n \, P. \tag{5.1}$$

It is essentially self-adjoint on $\mathcal{S}(\mathbb{R})$; denote its self-adjoint closure by G_n. The corresponding unitary displacement is

$$W_n = \exp(i G_n / \hbar). \tag{5.2}$$

Proposition 5.2 (Arithmetic commutator)**.** *For all $m, n \ge 1$, the restrictions of G_m and G_n to $\mathcal{S}(\mathbb{R})$ satisfy*

$$[G_m, G_n] = i\hbar (x_m y_n - y_m x_n) I \tag{5.3}$$

$$= i\hbar \ell \kappa \big(\log m \, \Omega(n) - \Omega(m) \log n \big) I. \tag{5.4}$$

Proof. Expand the commutator and use $[Q, P] = i\hbar I$. Terms involving $[Q, Q]$ and $[P, P]$ vanish. □

Theorem 5.3 (Arithmetic Heisenberg uncertainty law)**.** *For every normalized $\psi \in \mathcal{S}(\mathbb{R})$ and all $m, n \ge 1$,*

$$\boxed{\Delta_\psi G_m \, \Delta_\psi G_n \ge \frac{\hbar \ell \kappa}{2} \left| \Omega(m) \log n - \log m \, \Omega(n) \right|.} \tag{5.5}$$

For distinct primes p, q,

$$\boxed{\Delta_\psi G_p \, \Delta_\psi G_q \ge \frac{\hbar \ell \kappa}{2} \left| \log \frac{q}{p} \right|.} \tag{5.6}$$

For two powers of the same prime the lower bound is zero.

Proof. Apply the Robertson inequality to the self-adjoint pair G_m, G_n:

$$\Delta_\psi G_m \, \Delta_\psi G_n \ge \frac{1}{2} \left| \langle \psi, [G_m, G_n] \psi \rangle \right|. \tag{5.7}$$

Substitution of (5.4) gives (5.5). If p, q are distinct primes then $\Omega(p) = \Omega(q) = 1$, which gives (5.6). If $m = p^a, n = p^b$, the determinant $\Omega(m) \log n - \log m \, \Omega(n)$ vanishes. □

Remark 5.4 (The common invariant). Define the arithmetic symplectic area

$$\mathcal{A}(m,n) = \Omega(m)\log n - \log m\,\Omega(n). \tag{5.8}$$

The uncertainty lower bound is proportional to $|\mathcal{A}(m,n)|$, while the twisted product below carries the phase $\exp(i\vartheta\mathcal{A}(m,n)/2)$. Thus uncertainty and twisted Möbius inversion are controlled by the same arithmetic invariant, although they are distinct consequences of the CCR.

5.1. **The semiclassical and arithmetic-commutative limits are distinct.** The parameter controlling the Weyl cocycle is

$$\vartheta(\hbar,\ell,\kappa) = \frac{\ell\kappa}{\hbar}. \tag{5.9}$$

The uncertainty scale and the deformation scale therefore depend on the same three parameters in different ways.

Proposition 5.5 (Opposite fixed-scale limits)**.** *Fix $\ell > 0$ and $\kappa > 0$. For every fixed pair m, n,*

$$\frac{\hbar\ell\kappa}{2}|\mathcal{A}(m,n)| \longrightarrow 0 \qquad (\hbar \to 0^+), \tag{5.10}$$

whereas

$$|\vartheta| = \frac{\ell\kappa}{\hbar} \longrightarrow \infty. \tag{5.11}$$

Consequently the ordinary semiclassical limit $\hbar \to 0^+$ with ℓ, κ fixed does not approach the untwisted Dirichlet product.

Proof. Equation (5.10) follows directly from Theorem 5.3, while (5.11) follows from (5.9). □

Proposition 5.6 (Joint semiclassical–commutative scaling)**.** *Let $\ell = \ell(\hbar) > 0$ and $\kappa = \kappa(\hbar) > 0$. Then*

$$\vartheta(\hbar) \longrightarrow 0 \qquad (\hbar \to 0^+) \tag{5.12}$$

if and only if

$$\ell(\hbar)\kappa(\hbar) = o(\hbar). \tag{5.13}$$

Under this scaling the twisted product converges coefficientwise to the ordinary Dirichlet product, and the arithmetic uncertainty lower bound obeys

$$\frac{\hbar\ell(\hbar)\kappa(\hbar)}{2}|\mathcal{A}(m,n)| = o(\hbar^2) \tag{5.14}$$

for every fixed m, n.

Proof. The equivalence of (5.12) and (5.13) is immediate from (5.9). For fixed m, n, $\chi_\vartheta(m,n) \to 1$ as $\vartheta \to 0$, so the finite divisor sum defining $f \star_\vartheta g$ converges to ordinary Dirichlet convolution coefficientwise. Multiplying $\ell(\hbar)\kappa(\hbar) = o(\hbar)$ by $\hbar/2$ gives (5.14). □

Remark 5.7 (Two independent notions of classicality). The limit $\hbar \to 0^+$ suppresses the Robertson–Heisenberg lower bound, whereas $\vartheta \to 0$ removes the arithmetic twist. They are not the same operation. With fixed ℓ, κ they move in opposite directions; simultaneous suppression requires the joint scaling (5.13).

6. The Heisenberg-twisted Dirichlet algebra

6.1. Definition and associativity.

Definition 6.1 (Heisenberg-twisted Dirichlet convolution)**.** For arithmetic functions $f, g : \mathbb{N} \to \mathbb{C}$, define

$$(f \star_\vartheta g)(k) = \sum_{d|k} f(d)g(k/d)\chi_\vartheta(d, k/d). \tag{6.1}$$

The sum is finite for every k.

Lemma 6.2 (Cocycle identity)**.** *For all $a, b, c \in \mathbb{N}$,*

$$\chi_\vartheta(a, b)\chi_\vartheta(ab, c) = \chi_\vartheta(b, c)\chi_\vartheta(a, bc). \tag{6.2}$$

Proof. The exponent in (3.21) is the alternating bilinear form obtained from the additive map $n \mapsto z_n$. Expanding both sides using $\log(ab) = \log a + \log b$ and $\Omega(ab) = \Omega(a) + \Omega(b)$ gives the same exponent. □

Proposition 6.3 (Associative noncommutative arithmetic algebra)**.** *The product $\star_\vartheta$ is associative and has identity δ_1, where*

$$\delta_1(1) = 1, \qquad \delta_1(n) = 0 \quad (n > 1). \tag{6.3}$$

For $\vartheta \neq 0$ it is generally noncommutative.

Proof. Associativity follows from Lemma 6.2 after grouping the finite divisor sums by triples $abc = k$. The identity property follows from $\chi_\vartheta(1, n) = \chi_\vartheta(n, 1) = 1$. Noncommutativity follows from Corollary 3.10; equivalently,

$$(\delta_p \star_\vartheta \delta_q)(pq) = \chi_\vartheta(p, q) \tag{6.4}$$

whereas

$$(\delta_q \star_\vartheta \delta_p)(pq) = \chi_\vartheta(q, p) = \overline{\chi_\vartheta(p, q)}. \tag{6.5}$$

□

6.2. Weighted Banach algebra and operator representation.

Definition 6.4 (Weighted arithmetic Banach space)**.** For $\sigma > 0$, set

$$\mathcal{A}_\sigma = \left\{ f : \mathbb{N} \to \mathbb{C} : \|f\|_\sigma := \sum_{n\geq 1} \frac{|f(n)|}{n^\sigma} < \infty \right\}. \tag{6.6}$$

Proposition 6.5 (Banach-algebra estimate)**.** *For $f, g \in \mathcal{A}_\sigma$,*

$$\|f \star_\vartheta g\|_\sigma \leq \|f\|_\sigma \|g\|_\sigma. \tag{6.7}$$

Hence $(\mathcal{A}_\sigma, \star_\vartheta)$ is a unital Banach algebra.

Proof. Since $|\chi_\vartheta| = 1$,

$$\|f \star_\vartheta g\|_\sigma \leq \sum_{k\geq 1} \frac{1}{k^\sigma} \sum_{d|k} |f(d)||g(k/d)| \tag{6.8}$$

$$= \sum_{d,e\geq 1} \frac{|f(d)|}{d^\sigma} \frac{|g(e)|}{e^\sigma} \tag{6.9}$$

$$= \|f\|_\sigma \|g\|_\sigma. \tag{6.10}$$

Completeness is the completeness of weighted ℓ^1. □

Definition 6.6 (Weyl–Dirichlet representation)**.** For $f \in \mathcal{A}_\sigma$, define

$$\Pi_{\sigma,\vartheta}(f) = \sum_{n\geq 1} \frac{f(n)}{n^\sigma} W_n, \tag{6.11}$$

where the series converges absolutely in operator norm.

Proposition 6.7 (Representation theorem)**.** *For $f, g \in \mathcal{A}_\sigma$,*

$$\|\Pi_{\sigma,\vartheta}(f)\| \leq \|f\|_\sigma, \tag{6.12}$$

$$\Pi_{\sigma,\vartheta}(f)\Pi_{\sigma,\vartheta}(g) = \Pi_{\sigma,\vartheta}(f \star_\vartheta g). \tag{6.13}$$

Proof. The norm bound follows from $\|W_n\| = 1$. Absolute norm convergence permits multiplication and regrouping of the double series. The coefficient of W_k is exactly (6.1) by Theorem 3.9. □

7. Möbius inversion deformed by the CCR

7.1. The Möbius operator.

Definition 7.1 (Möbius operator)**.** For $\sigma > 1$, define

$$\mathcal{M}_{\sigma,\vartheta} = \Pi_{\sigma,\vartheta}(\mu) = \sum_{n\geq 1} \frac{\mu(n)}{n^\sigma} W_n. \tag{7.1}$$

The series converges absolutely in operator norm, so $\mathcal{M}_{\sigma,\vartheta} \in \mathcal{B}(\mathcal{H})$ and

$$\mathcal{D}(\mathcal{M}_{\sigma,\vartheta}) = \mathcal{H}. \tag{7.2}$$

Remark 7.2 (Concrete coordinate action)*.* For $\psi \in L^2(\mathbb{R})$,

$$(\mathcal{M}_{\sigma,\vartheta}\psi)(q) = \sum_{n\geq 1} \frac{\mu(n)}{n^\sigma} \exp\left[\frac{i}{\hbar} y_n \left(q - \frac{x_n}{2}\right)\right] \psi(q - x_n), \tag{7.3}$$

with convergence in L^2. Thus the operator is a Möbius-weighted superposition of simultaneous translations in position and momentum; it is not a Fourier multiplier when $\kappa \neq 0$.

7.2. Formal twisted inverse.

Proposition 7.3 (Existence and uniqueness of arithmetic inverses)**.** *Let $f : \mathbb{N} \to \mathbb{C}$ satisfy $f(1) \neq 0$. Then there exists a unique arithmetic function g such that*

$$f \star_\vartheta g = g \star_\vartheta f = \delta_1. \tag{7.4}$$

Its coefficients are determined recursively from $g(1) = f(1)^{-1}$.

Proof. For $n > 1$, the equation $(f \star_\vartheta r)(n) = 0$ contains the term $f(1)r(n)$ and otherwise only values $r(d)$ with $d < n$. It therefore constructs a unique right inverse r satisfying $f \star_\vartheta r = \delta_1$. Separately, the equations $(\lambda \star_\vartheta f)(n) = 0$ construct a unique left inverse λ satisfying $\lambda \star_\vartheta f = \delta_1$. Associativity then gives

$$\lambda = \lambda \star_\vartheta (f \star_\vartheta r) = (\lambda \star_\vartheta f) \star_\vartheta r = r, \tag{7.5}$$

so the inverse is bilateral and unique. □

Definition 7.4 (Heisenberg-zeta coefficients)**.** Let $z_\vartheta : \mathbb{N} \to \mathbb{C}$ be the unique two-sided inverse of μ for $\star_\vartheta$:

$$\mu \star_\vartheta z_\vartheta = z_\vartheta \star_\vartheta \mu = \delta_1. \tag{7.6}$$

Thus

$$z_\vartheta(1) = 1, \tag{7.7}$$

and for $n > 1$,

$$z_\vartheta(n) = -\sum_{\substack{d|n \\ d>1}} \mu(d) z_\vartheta(n/d) \chi_\vartheta(d, n/d). \tag{7.8}$$

Proposition 7.5 (Commuting boundary)**.** *At $\vartheta = 0$,*

$$z_0(n) = 1 \tag{7.9}$$

for every $n \geq 1$.

Proof. When $\vartheta = 0$, $\chi_0 \equiv 1$ and $\star_\vartheta$ is ordinary Dirichlet convolution. Equation (7.6) becomes $\mu * z_0 = \delta_1$. Classical Möbius inversion gives $z_0 = \mathbf{1}$. See [4]. □

Proposition 7.6 (Ordered-factorization formula)**.** *For $n > 1$, the twisted inverse has the finite coefficient expansion*

$$\begin{aligned} z_\vartheta(n) = \sum_{r\geq 1} (-1)^r \sum_{\substack{n_1 \cdots n_r = n \\ n_j > 1}} \left(\prod_{j=1}^{r} \mu(n_j) \right) \\ \times \exp\left[\frac{i\vartheta}{2} \sum_{1 \leq i < j \leq r} \left(\Omega(n_i) \log n_j - \log n_i \, \Omega(n_j) \right) \right]. \end{aligned} \tag{7.10}$$

Only finitely many r and ordered factorizations contribute for fixed n.

Proof. Write $\mu = \delta_1 + h$ with $h(1) = 0$. Coefficientwise the formal Neumann expansion is

$$z_\vartheta = \sum_{r \geq 0} (-h)^{\star_\vartheta r}. \tag{7.11}$$

For fixed n, a product of more than $\log_2 n$ factors exceeding one cannot equal n, so the coefficient sum is finite. Iterating the bicharacter cocycle yields the pairwise phase in (7.10). □

7.3. **A bounded inverse in a rigorous half-plane.**

Definition 7.7 (Uniform Neumann threshold)**.** Let σ_* be the unique real number > 1 satisfying

$$\frac{\zeta(\sigma_*)}{\zeta(2\sigma_*)} = 2. \tag{7.12}$$

Numerically,

$$\sigma_* = 1.578017063034\ldots. \tag{7.13}$$

Theorem 7.8 (Uniform Neumann region)**.** *For every real ϑ and every $\sigma > \sigma_*$,*

$$z_\vartheta \in \mathcal{A}_\sigma. \tag{7.14}$$

Consequently

$$\mathcal{Z}_{\sigma,\vartheta} = \Pi_{\sigma,\vartheta}(z_\vartheta) = \sum_{n \geq 1} \frac{z_\vartheta(n)}{n^\sigma} W_n \tag{7.15}$$

converges absolutely in operator norm and satisfies

$$\mathcal{M}_{\sigma,\vartheta} \mathcal{Z}_{\sigma,\vartheta} = \mathcal{Z}_{\sigma,\vartheta} \mathcal{M}_{\sigma,\vartheta} = I. \tag{7.16}$$

Proof. Write $\mu = \delta_1 + h$, where $h(1) = 0$. For $\sigma > 1$,

$$\|h\|_\sigma = \sum_{n\geq 2} \frac{\mu(n)^2}{n^\sigma} \tag{7.17}$$

$$= \frac{\zeta(\sigma)}{\zeta(2\sigma)} - 1. \tag{7.18}$$

The right-hand side is continuous and strictly decreasing from $+\infty$ to 0 as σ runs from 1^+ to $+\infty$, so σ_* is unique and $\|h\|_\sigma < 1$ exactly when $\sigma > \sigma_*$. The Banach-algebra Neumann series

$$(\delta_1 + h)^{-1} = \sum_{r=0}^{\infty} (-h)^{\star_\vartheta r} \tag{7.19}$$

then converges in $\mathcal{A}_\sigma$. Uniqueness identifies it with z_ϑ, and the representation theorem gives (7.16). □

Warning 7.9 (Exact border of the proved inverse). The Möbius operator exists as a bounded operator for every $\sigma > 1$. The argument above proves a bounded inverse only for $\sigma > \sigma_*$. At $\sigma = \sigma_*$ the Neumann norm equals one and this argument is inconclusive. No claim is made here for $1 < \sigma \leq \sigma_*$.

8. The first Heisenberg arithmetic law

Theorem 8.1 (Prime rays are rigid; distinct primes interfere)**.** *Let p, q be distinct primes and let $r \geq 1$. Then*

$$z_\vartheta(p^r) = 1, \tag{8.1}$$

$$z_\vartheta(pq) = 2\cos\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) - 1, \tag{8.2}$$

$$z_\vartheta(pq) - 1 = -4\sin^2\left(\frac{\vartheta}{4}\log\frac{q}{p}\right). \tag{8.3}$$

Proof. For p^r, only the divisors 1 and p contribute to the recursion because $\mu(p^j) = 0$ for $j \geq 2$. Moreover,

$$\Omega(p)\log(p^{r-1}) - \log p\,\Omega(p^{r-1}) = 0, \tag{8.4}$$

so $\chi_\vartheta(p, p^{r-1}) = 1$. Equation (7.8) gives

$$z_\vartheta(p^r) = z_\vartheta(p^{r-1}), \tag{8.5}$$

and (8.1) follows from $z_\vartheta(1) = 1$.

For pq, the four divisors $1, p, q, pq$ give

$$0 = z_\vartheta(pq) - \chi_\vartheta(p, q) - \chi_\vartheta(q, p) + 1. \tag{8.6}$$

Since

$$\chi_\vartheta(p, q) = \exp\left(\frac{i\vartheta}{2}\log\frac{q}{p}\right), \tag{8.7}$$

and $\chi_\vartheta(q, p)$ is its complex conjugate, (8.2) follows. The identity (8.3) is the elementary formula $2(\cos u - 1) = -4\sin^2(u/2)$. □

Corollary 8.2 (Loss of ordinary multiplicativity)**.** *Generically,*

$$z_\vartheta(pq) \neq z_\vartheta(p) z_\vartheta(q) = 1. \tag{8.8}$$

Thus the Heisenberg deformation preserves every one-prime ray but destroys ordinary multiplicativity through cross-prime interference.

Corollary 8.3 (Small-area expansion)**.** *For fixed distinct primes p, q,*

$$z_\vartheta(pq) = 1 - \frac{\vartheta^2}{4}\left(\log\frac{q}{p}\right)^2 + O(\vartheta^4) \qquad (\vartheta \to 0). \tag{8.9}$$

There is no linear correction.

Proposition 8.4 (Two-prime lattice recurrence)**.** *Let $p \neq q$ be primes, set*

$$Z_{a,b} = z_\vartheta(p^a q^b), \qquad \lambda = \frac{\vartheta}{2}\log\frac{q}{p}. \tag{8.10}$$

Then for $a, b \geq 1$,

$$Z_{a,b} = e^{ib\lambda} Z_{a-1,b} + e^{-ia\lambda} Z_{a,b-1} - e^{i(b-a)\lambda} Z_{a-1,b-1}, \tag{8.11}$$

with boundary values

$$Z_{a,0} = Z_{0,b} = 1. \tag{8.12}$$

In particular,

$$z_\vartheta(p^2 q) = 1 + 2\cos(2\lambda) - 2\cos\lambda. \tag{8.13}$$

Proof. Apply the right-inverse recursion (7.8) to $p^a q^b$. Since $\mu(d) \neq 0$ only when the exponents of p, q in d are at most one, only $d = p, q, pq$ contribute. Evaluation of the three cocycle factors gives (8.11); the boundary values are Theorem 8.1 on prime rays. □

Proposition 8.5 (Evenness of the full inverse)**.** *For every $n \geq 1$,*

$$z_{-\vartheta}(n) = z_\vartheta(n). \tag{8.14}$$

Proof. Changing ϑ to $-\vartheta$ reverses the cocycle:

$$(f \star_{-\vartheta} g)(n) = (g \star_\vartheta f)(n). \tag{8.15}$$

Since z_ϑ is a two-sided inverse of μ for $\star_\vartheta$, it is a two-sided inverse of μ for the opposite product $\star_{-\vartheta}$. Uniqueness of the inverse gives (8.14). □

Proposition 8.6 (Reality for real deformation)**.** *For real ϑ and every $n \geq 1$,*

$$z_\vartheta(n) \in \mathbb{R}. \tag{8.16}$$

Proof. Complex conjugation changes χ_ϑ to $\chi_{-\vartheta}$. Since μ is real, conjugating the inverse equation shows that $\overline{z_\vartheta}$ is the inverse for the $-\vartheta$ product. Proposition 8.5 therefore gives $\overline{z_\vartheta} = z_{-\vartheta} = z_\vartheta$. □

Remark 8.7 (Why this is the non-tautological step). At $\vartheta = 0$ the inverse coefficients are identically 1. At nonzero ϑ, equation (8.3) creates a new arithmetic coefficient depending on the logarithmic separation of two primes. This dependence is forced by the symplectic area in the Weyl product and disappears if the CCR is removed. It is therefore the first result in the notebook where Heisenberg noncommutativity changes the arithmetic inverse itself rather than merely constraining an unrelated state space.

9. Cohomological curvature of the twisted Möbius inverse

Set

$$\mathcal{A}(d, e) = \Omega(d)\log e - \log d\,\Omega(e), \tag{9.1}$$

so that

$$\chi_\vartheta(d, e) = \exp\left(\frac{i\vartheta}{2}\mathcal{A}(d, e)\right). \tag{9.2}$$

Proposition 9.1 (Coefficientwise entire dependence)**.** *For each fixed $n \geq 1$, the map $\vartheta \mapsto z_\vartheta(n)$ is entire. It is even, and it is real-valued on the real axis.*

Proof. Formula (7.10) is a finite sum for fixed n, each term being an exponential $e^{ic\vartheta}$. Evenness and reality follow from Propositions 8.5 and 8.6. □

Proposition 9.2 (Second derivative of the inverse)**.** *Define*

$$g(k) = \frac{1}{4}\sum_{d|k} \mu(d)\mathcal{A}(d, k/d)^2. \tag{9.3}$$

Then

$$\boxed{z_0''(n) = \sum_{k|n} g(k).} \tag{9.4}$$

Proof. Differentiate the inverse equation

$$0 = \sum_{d|n} \mu(d) z_\vartheta(n/d) \exp\Big(\frac{i\vartheta}{2}\mathcal{A}(d, n/d)\Big) \tag{9.5}$$

twice at $\vartheta = 0$. Since $z_0 \equiv 1$ and $z_0' \equiv 0$,

$$(\mu * z_0'')(n) = \frac{1}{4}\sum_{d|n} \mu(d)\mathcal{A}(d, n/d)^2 = g(n). \tag{9.6}$$

Convolving with $\mathbf{1} = \mu^{-1}$ gives (9.4). □

Lemma 9.3 (Support collapse)**.** *Let $k = \prod_{j=1}^r p_j^{a_j}$ with distinct primes p_j. Then*

$$g(k) = 0 \qquad \text{for } r = 0,\ r = 1,\ \text{or } r \geq 3. \tag{9.7}$$

For two distinct primes,

$$\boxed{g(p^\alpha q^\beta) = -\frac{\alpha\beta}{2}\log^2\frac{q}{p}.} \tag{9.8}$$

Proof. A contributing divisor is square-free and therefore corresponds to a subset $S \subseteq \{1, \dots, r\}$. Put

$$R = \Omega(k) = \sum_j a_j, \qquad L = \log k, \qquad X_S = \sum_{j\in S} \log p_j. \tag{9.9}$$

Then

$$\mathcal{A}(d, k/d) = |S|L - RX_S. \tag{9.10}$$

Its square has Boolean degree at most two, so its full alternating difference vanishes when $r \geq 3$. For $r = 1$ the area is zero. For $r = 2$, direct evaluation of the four subsets gives

$$\sum_{d|p^\alpha q^\beta} \mu(d)\mathcal{A}(d, p^\alpha q^\beta/d)^2 = -2\alpha\beta\log^2(q/p), \tag{9.11}$$

which yields (9.8). □

Theorem 9.4 (Exact cohomological curvature)**.** *If*

$$n = \prod_{j=1}^r p_j^{a_j}, \qquad T(a) = \frac{a(a+1)}{2}, \tag{9.12}$$

then

$$\boxed{z_0''(n) = -\frac{1}{2}\sum_{1\leq i<j\leq r} T(a_i)T(a_j)\log^2\frac{p_j}{p_i}.} \tag{9.13}$$

Thus $-z_0''(n) \geq 0$, with equality exactly for $n = 1$ or a prime power.

Proof. Only divisors supported on exactly two primes survive by Lemma 9.3. Summing (9.8) over $1 \le \alpha \le a_i$, $1 \le \beta \le a_j$ gives the factor $T(a_i)T(a_j)$. □

Corollary 9.5 (Square-free prime-log variance)**.** *If $n = p_1 \cdots p_r$ is square-free, then*

$$-z_0''(n) = \frac{1}{2}\sum_{i<j}(\log p_j - \log p_i)^2 \tag{9.14}$$

$$= \frac{1}{2}\left[r\sum_{p|n}(\log p)^2 - (\log n)^2\right] \tag{9.15}$$

$$= \frac{r^2}{2}\,\mathrm{Var}_{p|n}(\log p). \tag{9.16}$$

Corollary 9.6 (Infinitesimal cohomological size)**.** *For square-free $n = p_1 \cdots p_r$, let*

$$B_{ij} = \log p_j - \log p_i. \tag{9.17}$$

Then B has rank at most two. If $r \ge 2$, its two nonzero singular values are both

$$s(n) = \sqrt{r\sum_{p|n}(\log p)^2 - (\log n)^2}, \tag{9.18}$$

and

$$\boxed{-z_0''(n) = \frac{1}{2}s(n)^2.} \tag{9.19}$$

Proof. Write $B = \mathbf{1}a^T - a\mathbf{1}^T$, $a_i = \log p_i$. The two nonzero singular values of $uv^T - vu^T$ equal

$$\sqrt{\|u\|^2\|v\|^2 - (u\cdot v)^2}. \tag{9.20}$$

Apply this with $u = \mathbf{1}$, $v = a$, and then use Corollary 9.5. □

Corollary 9.7 (Exact Taylor law for each integer)**.** *For every fixed n,*

$$z_\vartheta(n) = 1 + \frac{\vartheta^2}{2}z_0''(n) + O_n(\vartheta^4), \tag{9.21}$$

and the Taylor series has infinite radius of convergence. For square-free n,

$$\boxed{z_\vartheta(n) = 1 - \frac{\vartheta^2}{4}\left[\omega(n)\sum_{p|n}(\log p)^2 - (\log n)^2\right] + O_n(\vartheta^4).} \tag{9.22}$$

10. Dirichlet summation of the curvature

Define

$$c(n) = -z_0''(n) \ge 0. \tag{10.1}$$

Theorem 10.1 (Exact curvature Dirichlet series)**.** *For $\Re s > 1$,*

$$\mathcal{K}(s) = \sum_{n\ge1}\frac{c(n)}{n^s} \tag{10.2}$$

converges absolutely and

$$\boxed{\mathcal{K}(s) = \frac{\zeta(s)}{2}\left[S_0(s)S_2(s) - S_1(s)^2\right],} \tag{10.3}$$

where

$$S_j(s) = \sum_p \frac{p^{-s}}{(1-p^{-s})^2} (\log p)^j, \qquad j = 0, 1, 2. \tag{10.4}$$

Equivalently,

$$S_0 S_2 - S_1^2 = \sum_{p<q} \frac{p^{-s} q^{-s}}{(1-p^{-s})^2 (1-q^{-s})^2} \log^2 \frac{q}{p}. \tag{10.5}$$

Proof. Use Theorem 9.4 and

$$\sum_{a \geq 1} T(a) x^a = \frac{x}{(1-x)^3}. \tag{10.6}$$

Factoring the unrestricted Euler factors gives $\zeta(s)$, while a selected prime contributes

$$\frac{p^{-s}}{(1-p^{-s})^2}. \tag{10.7}$$

The pair sum is exactly the weighted variance determinant $S_0 S_2 - S_1^2$. Absolute convergence follows for $\Re s > 1$. □

Corollary 10.2 (Prime-zeta reduction)**.** *Let $P(s) = \sum_p p^{-s}$ for $\Re s > 1$. Then*

$$S_0(s) = \sum_{m \geq 1} m P(ms), \tag{10.8}$$

$$S_1(s) = -\sum_{m \geq 1} m P'(ms), \tag{10.9}$$

$$S_2(s) = \sum_{m \geq 1} m P''(ms). \tag{10.10}$$

Using

$$P(s) = \sum_{k \geq 1} \frac{\mu(k)}{k} \log \zeta(ks), \tag{10.11}$$

gives

$$S_0(s) = \sum_{r \geq 1} \frac{J_2(r)}{r} \log \zeta(rs), \tag{10.12}$$

$$S_1(s) = -\sum_{r \geq 1} \varphi(r) \frac{\zeta'(rs)}{\zeta(rs)}, \tag{10.13}$$

$$S_2(s) = \left(\frac{\zeta'}{\zeta} \right)'(s), \tag{10.14}$$

where $J_2(r) = r^2 \prod_{p \mid r} (1 - p^{-2})$.

Proof. Expand $x/(1-x)^2 = \sum_{m\ge1} mx^m$, substitute (10.11), and group by $r = mk$. The needed divisor identities are

$$r\sum_{k|r}\frac{\mu(k)}{k^2} = \frac{J_2(r)}{r}, \tag{10.15}$$

$$r\sum_{k|r}\frac{\mu(k)}{k} = \varphi(r), \tag{10.16}$$

$$r\sum_{k|r}\mu(k) = \begin{cases}1, & r=1,\\ 0, & r>1.\end{cases} \tag{10.17}$$

The last identity collapses S_2 to (10.14). □

Proposition 10.3 (Singularity at $s=1$)**.** *As $s\to1^+$,*

$$\boxed{\mathcal{K}(s) \sim \frac{1}{2(s-1)^3}\log\frac{1}{s-1}.} \tag{10.18}$$

Proof. From (10.11),

$$P(s) = \log\frac{1}{s-1} + O(1). \tag{10.19}$$

Hence

$$S_0(s) = \log\frac{1}{s-1} + O(1), \quad S_1(s) = \frac{1}{s-1} + O(1), \quad S_2(s) = \frac{1}{(s-1)^2} + O(1). \tag{10.20}$$

Combine with $\zeta(s) \sim (s-1)^{-1}$. □

11. Continuation of the curvature and detection of zeta zeros

Theorem 11.1 (Local continuation to $\Re s > 1/2$)**.** *Let $U \subset \{\Re s > 1/2\}$ be simply connected and contain no zero or pole of $\zeta(s)$. Choose a branch of $\log\zeta(s)$ on U. Then* (10.12)–(10.14)*, substituted into* (10.3)*, continue $\mathcal{K}$ holomorphically to U.*

Proof. For $r \ge 2$, $\Re(rs) > 1$, so the corresponding terms are holomorphic and converge normally on compact subsets. The $r=1$ terms are holomorphic by the hypothesis on U. □

Theorem 11.2 (Zeta zeros are logarithmic curvature singularities)**.** *Let ρ be a zero of multiplicity $m \ge 1$ with $1/2 < \Re\rho < 1$, and write*

$$\zeta(s) = c_\rho(s-\rho)^m(1 + O(s-\rho)), \qquad c_\rho \neq 0. \tag{11.1}$$

Then

$$\boxed{\mathcal{K}(s) = -\frac{c_\rho m^2}{2}(s-\rho)^{m-2}\log(s-\rho) + O\Big((s-\rho)^{m-2}\Big).} \tag{11.2}$$

For a simple zero,

$$\boxed{\mathcal{K}(s) = -\frac{\zeta'(\rho)}{2}\frac{\log(s-\rho)}{s-\rho} + O\Big(\frac{1}{s-\rho}\Big).} \tag{11.3}$$

Proof. For $r \ge 2$, the terms in (10.12)–(10.13) are holomorphic near ρ. With $t = s-\rho$, the singular pieces are

$$S_0 = m\log t + O(1), \qquad S_1 = -\frac{m}{t} + O(1), \qquad S_2 = -\frac{m}{t^2} + O(1). \tag{11.4}$$

Thus

$$S_0S_2 - S_1^2 = -\frac{m^2 \log t}{t^2} + O(t^{-2}). \tag{11.5}$$

Multiply by $\zeta(s)/2$. □

Corollary 11.3 (Curvature criterion equivalent to RH)**.** *RH is equivalent to the absence of logarithmic branch singularities of $\mathcal{K}$ in*

$$\boxed{\frac{1}{2} < \Re s < 1.} \tag{11.6}$$

Proof. Theorem 11.2 detects every zero in the right half of the critical strip, while Theorem 11.1 gives holomorphy on zero-free neighbourhoods. The functional equation makes the absence of zeros in this half-strip equivalent to RH. □

12. The Heisenberg-deformed zeta series

Definition 12.1 (Heisenberg-deformed zeta series)**.** For $\Re s > \sigma_*$, define

$$\zeta_{\mathrm{H}}(s;\vartheta) = \sum_{n\geq 1} \frac{z_\vartheta(n)}{n^s}. \tag{12.1}$$

Theorem 7.8 guarantees absolute convergence for every $\Re s > \sigma_*$.

Proposition 12.2 (Classical boundary)**.** *For $\Re s > 1$,*

$$\zeta_{\mathrm{H}}(s;0) = \zeta(s). \tag{12.2}$$

Proof. By Proposition 7.5, $z_0(n) = 1$ for all n. Substitution into (12.1) gives the ordinary Dirichlet series for $\zeta(s)$. □

Proposition 12.3 (Even Heisenberg deformation)**.** *For $\Re s > \sigma_*$,*

$$\zeta_{\mathrm{H}}(s;-\vartheta) = \zeta_{\mathrm{H}}(s;\vartheta). \tag{12.3}$$

Proof. This follows termwise from Proposition 8.5 and absolute convergence. □

Warning 12.4 (No analytic continuation is asserted). The series (12.1) is absolutely convergent and holomorphic on the open half-plane $\Re s > \sigma_*$. No meromorphic continuation, functional equation, scalar Euler product, or critical-strip theory for $\zeta_{\mathrm{H}}(s;\vartheta)$ is asserted here.

Proposition 12.5 (Conditional first-order rigidity of simple zeros)**.** *Assume that, for some simple zero ρ of the ordinary zeta function, there exists a function $F(s,\vartheta)$ analytic in a neighbourhood of $(\rho, 0)$ such that*

$$F(s,0) = \zeta(s), \tag{12.4}$$

$$F(s,-\vartheta) = F(s,\vartheta), \tag{12.5}$$

and that F agrees with ζ_{H} wherever both are defined. Then the local zero branch $\rho(\vartheta)$ through ρ satisfies

$$\rho'(0) = 0. \tag{12.6}$$

Proof. The implicit-function theorem applies because $\zeta'(\rho) \neq 0$. Evenness (12.5) makes the unique local zero branch even, so its first derivative at the origin vanishes. □

Warning 12.6. Proposition 12.5 is conditional on an analytic continuation in both variables and does not supply such a continuation. It is included because it identifies a precise way in which the exact $\vartheta \mapsto -\vartheta$ symmetry would constrain the motion of classical Riemann zeros under the Heisenberg deformation if that continuation exists.

13. A genuine Möbius Hamiltonian

13.1. **Definition and domain.**

Definition 13.1 (Cosine and sine Weyl quadratures)**.** For $n \geq 1$, set

$$C_n = \frac{W_n + W_n^*}{2}, \tag{13.1}$$

$$J_n = \frac{W_n - W_n^*}{2i}. \tag{13.2}$$

Both are bounded self-adjoint operators with norm at most 1.

Definition 13.2 (Möbius Hamiltonian)**.** Fix an energy scale $E_0 > 0$ and $\sigma > 1$. Define

$$H_{\mu,\sigma,\vartheta} = E_0 \left(\operatorname{Re} \mathcal{M}_{\sigma,\vartheta} - I\right) = E_0 \sum_{n\geq 2} \frac{\mu(n)}{n^\sigma} C_n. \tag{13.3}$$

The series converges absolutely in operator norm.

Proposition 13.3 (Self-adjoint bounded dynamics)**.** *For $\sigma > 1$, $H_{\mu,\sigma,\vartheta}$ is bounded and self-adjoint on all of $\mathcal{H}$:*

$$\mathcal{D}(H_{\mu,\sigma,\vartheta}) = \mathcal{H}. \tag{13.4}$$

It therefore generates the unitary group

$$U(t) = \exp\left(-\frac{i}{\hbar} t H_{\mu,\sigma,\vartheta}\right). \tag{13.5}$$

Proof. The norm series in (13.3) is dominated by $E_0 \sum_{n\geq 2} |\mu(n)| n^{-\sigma}$, which converges for $\sigma > 1$. Every C_n is self-adjoint, hence so is the norm limit. The spectral theorem then defines (13.5). □

Proposition 13.4 (Explicit Schrödinger equation)**.** *For $\psi_0 \in \mathcal{H}$, the evolution $\psi(t) = U(t)\psi_0$ satisfies*

$$i\hbar\, \partial_t \psi(t,q) = (H_{\mu,\sigma,\vartheta}\psi(t))(q) \tag{13.6}$$

in the strong L^2 sense, where

$$(H_{\mu,\sigma,\vartheta}\psi)(q) = \frac{E_0}{2} \sum_{n\geq 2} \frac{\mu(n)}{n^\sigma} \Bigg[e^{\frac{i}{\hbar} y_n (q - x_n/2)} \psi(q - x_n) + e^{-\frac{i}{\hbar} y_n (q + x_n/2)} \psi(q + x_n) \Bigg]. \tag{13.7}$$

Proof. Equation (13.7) follows from Proposition 3.2 applied to W_n and W_n^*. The series converges in operator norm, so the bounded Hamiltonian gives a strong Schrödinger evolution for every initial vector. □

13.2. **Exact equations of motion for position and momentum.**

Lemma 13.5 (Bounded first commutators)**.** *For $\sigma > 1$, the series*

$$\sum_{n\geq 2} \frac{|\mu(n)| x_n}{n^\sigma}, \tag{13.8}$$

$$\sum_{n\geq 2} \frac{|\mu(n)| y_n}{n^\sigma} \tag{13.9}$$

converge. Hence $[H_{\mu,\sigma,\vartheta}, Q]$ and $[H_{\mu,\sigma,\vartheta}, P]$, initially defined on $\mathcal{S}(\mathbb{R})$, extend to bounded operators.

Proof. The first series is bounded by a constant multiple of $\sum n^{-\sigma}\log n$. Since

$$\Omega(n) \le \frac{\log n}{\log 2}, \tag{13.10}$$

the second is bounded by another constant multiple of the same convergent series. □

Theorem 13.6 (Exact Heisenberg equations of motion)**.** *Let*

$$Q(t) = U(t)^* Q U(t), \tag{13.11}$$

$$P(t) = U(t)^* P U(t). \tag{13.12}$$

The bounded commutators in Lemma 13.5 imply that $U(t)$ preserves $\mathcal{D}(Q)$ and $\mathcal{D}(P)$. On these domains, and equivalently as quadratic-form identities, the Heisenberg equations are

$$\dot{Q}(t) = \frac{E_0}{\hbar} U(t)^* \left[\sum_{n\ge 2} \frac{\mu(n) x_n}{n^\sigma} J_n \right] U(t), \tag{13.13}$$

$$\dot{P}(t) = \frac{E_0}{\hbar} U(t)^* \left[\sum_{n\ge 2} \frac{\mu(n) y_n}{n^\sigma} J_n \right] U(t). \tag{13.14}$$

At $t = 0$ this becomes

$$\dot{Q}(0) = \frac{E_0 \ell}{\hbar} \sum_{n\ge 2} \frac{\mu(n)\log n}{n^\sigma} J_n, \tag{13.15}$$

$$\dot{P}(0) = \frac{E_0 \kappa}{\hbar} \sum_{n\ge 2} \frac{\mu(n)\Omega(n)}{n^\sigma} J_n. \tag{13.16}$$

Proof. For $A = Q$ or P, the bounded-commutator theorem for unitary dynamics gives

$$\dot{A}(t) = \frac{i}{\hbar} U(t)^* [H_{\mu,\sigma,\vartheta}, A] U(t). \tag{13.17}$$

From Proposition 3.4,

$$[Q, C_n] = i x_n J_n, \tag{13.18}$$

$$[P, C_n] = i y_n J_n. \tag{13.19}$$

Therefore

$$[H_{\mu,\sigma,\vartheta}, Q] = -iE_0 \sum_{n\ge 2} \frac{\mu(n) x_n}{n^\sigma} J_n, \tag{13.20}$$

$$[H_{\mu,\sigma,\vartheta}, P] = -iE_0 \sum_{n\ge 2} \frac{\mu(n) y_n}{n^\sigma} J_n. \tag{13.21}$$

Lemma 13.5 justifies the sums as bounded operators. Substitution into (13.17) gives (13.13) and (13.14). □

Remark 13.7 (Physical and arithmetic meaning of the two equations). The position variable is Q, the logarithmic-scale coordinate. The momentum variable is $P = -i\hbar\partial_q$, the generator conjugate to that coordinate. Multiplication by an integer n is represented by a phase-space displacement of size $\ell \log n$ in Q and $\kappa\Omega(n)$ in P. Equations (13.15) and (13.16) therefore say that the Möbius Hamiltonian drives position by logarithmic size and momentum by prime-factor count, with the same quantum current operators J_n.

13.3. **The CCR is visible in the dynamics.**

Proposition 13.8 (Pairwise dynamical noncommutativity)**.** *Extend* log *and* Ω *additively to positive rationals and define* W_r *by* (3.19) *for* $r \in \mathbb{Q}_{>0}$*. For* $m, n \in \mathbb{N}$*, put*

$$\Phi_{m,n} = \vartheta(\Omega(m)\log n - \log m\,\Omega(n)). \tag{13.22}$$

Then

$$[C_m, J_n] = \frac{1}{2}\sin\Big(\frac{\Phi_{m,n}}{2}\Big)\Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big). \tag{13.23}$$

Proof. Expand C_m and J_n using (13.1)–(13.2). Apply (3.6) to the four pairs (z_m, z_n), $(z_m, -z_n)$, $(-z_m, z_n)$, and $(-z_m, -z_n)$. □

Corollary 13.9 (Second-order arithmetic commutator)**.** *At* $t = 0$*,*

$$\ddot{Q}(0) = \frac{iE_0^2}{2\hbar^2}\sum_{m,n\geq 2}\frac{\mu(m)\mu(n)x_n}{(mn)^\sigma}\sin\Big(\frac{\Phi_{m,n}}{2}\Big) \times \Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big), \tag{13.24}$$

and the analogous momentum equation is

$$\ddot{P}(0) = \frac{iE_0^2}{2\hbar^2}\sum_{m,n\geq 2}\frac{\mu(m)\mu(n)y_n}{(mn)^\sigma}\sin\Big(\frac{\Phi_{m,n}}{2}\Big) \times \Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big). \tag{13.25}$$

The double sums converge in operator norm for $\sigma > 1$*.*

Proof. Differentiate (13.13) and (13.14) once more and use Proposition 13.8. Polynomial logarithmic weights are summable against $(mn)^{-\sigma}$ for $\sigma > 1$. □

Corollary 13.10 (The interaction disappears with the Weyl cocycle)**.** *If* $\vartheta = 0$*, then*

$$\Phi_{m,n} = 0 \tag{13.26}$$

for all m, n*, and all pairwise terms in* (13.24)–(13.25) *vanish. Thus the second-order cross-arithmetic second-order term displayed above is generated entirely by the exact Heisenberg cocycle.*

14. Exact non-localization inside the arithmetic dynamics

Proposition 14.1 (Schwartz action and distributional transpose)**.** *For* $\sigma > 1$*, the Hamiltonian* $H_{\mu,\sigma,\vartheta}$ *maps* $\mathcal{S}(\mathbb{R})$ *continuously into itself. Its continuous transpose* $H^\times : \mathcal{S}'(\mathbb{R}) \to \mathcal{S}'(\mathbb{R})$ *is defined by*

$$\langle H^\times T, \phi\rangle = \langle T, H_{\mu,\sigma,\vartheta}\phi\rangle, \qquad T \in \mathcal{S}'(\mathbb{R}), \quad \phi \in \mathcal{S}(\mathbb{R}). \tag{14.1}$$

Then $H^\times$ *is continuous on* $\mathcal{S}'(\mathbb{R})$*.*

Proof. For every Schwartz seminorm $p_{a,b}(\phi) = \sup_q |q^a\phi^{(b)}(q)|$, the coordinate action of a Weyl operator gives

$$p_{a,b}(W_n\phi) \leq C_{a,b}(1 + |x_n| + |y_n|)^{N_{a,b}}\sum_{j\leq b} p_{a+j,b-j}(\phi) \tag{14.2}$$

for suitable constants. Since $x_n, y_n = O(\log n)$ and

$$\sum_{n\geq 2} n^{-\sigma}(1 + \log n)^N < \infty \qquad (\sigma > 1) \tag{14.3}$$

for every fixed N, the Hamiltonian series converges in every Schwartz seminorm and defines a continuous map $\mathcal{S} \to \mathcal{S}$. Therefore (14.1) defines the standard continuous dual action on $\mathcal{S}'$. No

termwise extension to all tempered distributions is assumed without this transpose construction. □

Proposition 14.2 (No exact position point is a stationary generalized mode)**.** *Let $\sigma > 1$, $\ell > 0$, and $E_0 > 0$. For no $q_0 \in \mathbb{R}$ and no scalar $E \in \mathbb{C}$ does $\delta_{q_0} \in \mathcal{S}'(\mathbb{R})$ satisfy*

$$H_{\mu,\sigma,\vartheta}^{\times}\delta_{q_0} = E\delta_{q_0}. \tag{14.4}$$

Proof. The transpose action of a Weyl displacement on a point mass is

$$W(x,y)^{\times}\delta_{q_0} = \exp\left(\frac{i}{\hbar}y(q_0 + x/2)\right)\delta_{q_0+x}, \tag{14.5}$$

$$(W(x,y)^{*})^{\times}\delta_{q_0} = \exp\left(-\frac{i}{\hbar}y(q_0 - x/2)\right)\delta_{q_0-x}. \tag{14.6}$$

The $n = 2$ term therefore contributes nonzero masses at $q_0 \pm \ell \log 2$. No other positive integer gives the positive shift $\ell \log 2$, and no adjoint term gives that same positive shift. Since $\mu(2) = -1$, the corresponding coefficient cannot cancel. Hence the support of the left side of (14.4) contains a point distinct from q_0, while the right side is supported only at q_0. □

Remark 14.3 (Relation to exact Heisenberg impossibility)*.* Theorem 2.9 is a universal consequence of the CCR. Proposition 14.2 is more specific: the Möbius Hamiltonian itself does not admit an exactly position-localized stationary mode. The reason is not a measurement uncertainty but the exact translation content of the arithmetic Weyl operators.

15. Status classification in Version 4

15.1. **Standard ingredients.** The following ingredients are standard and are not presented as new:

(1) the Schrödinger representation of Q and P and the CCR (2.7);
(2) Weyl operators and the Weyl product (3.5);
(3) the arithmetic functions μ, Ω, and classical Möbius inversion;
(4) weighted ℓ^1 Banach-algebra estimates and Neumann inversion.

15.2. **Original notebook constructions.** Version 4 adds:

(V4.1) the quadratic-obstruction theorem Theorem 4.1;
(V4.2) coefficientwise entire dependence of $z_\vartheta(n)$;
(V4.3) the exact curvature law (9.13);
(V4.4) the square-free prime-log variance identity (9.16);
(V4.5) the curvature Dirichlet determinant (10.3);
(V4.6) the identity $S_2 = (\zeta'/\zeta)'$;
(V4.7) the logarithmic zero singularity (11.2);
(V4.8) the curvature criterion Corollary 11.3.

The following are the constructions/results introduced in this notebook and are the candidates for substantive originality:

(1) the arithmetic phase-space embedding $n \mapsto (\ell \log n, \kappa\Omega(n))$ coupled to the Weyl representation;
(2) the cocycle (3.21) as a deformation of Dirichlet convolution;
(3) the arithmetic uncertainty law (5.5), which specializes the Heisenberg–Robertson inequality to the integer generators G_n;
(4) the two-sided Heisenberg-twisted Möbius inverse z_ϑ;
(5) the exact prime-ray/prime-pair law (8.1)–(8.3);
(6) the evenness theorem (8.14);
(7) the Möbius Hamiltonian (13.3) and its exact position–momentum equations (13.13)–(13.14);
(8) the pairwise second-order commutator controlled by $\sin(\Phi_{m,n}/2)$.

16. Corrections and extensions carried into Version 4

Remark 16.1. Version 4 incorporates the following mathematical corrections and clarifications: the cocycle is proved nontrivial rather than merely described as noncommutative; no faithfulness of $\Pi_{\sigma,\vartheta}$ is claimed; left and right inverses are constructed separately before being identified; the uniform Neumann half-plane is sharpened to $\sigma > \sigma_*$; the ordered-factorization formula and a two-prime recurrence are included; real-valuedness of $z_\vartheta(n)$ for real ϑ is proved; the deformed Dirichlet series is asserted holomorphic only on its proved open half-plane; the unitary evolution is not claimed to preserve Schwartz space; and the action on tempered distributions is defined by continuous transposition.

17. Open problems

Question 17.1 (Analytic continuation)**.** Does $\zeta_{\mathrm{H}}(s;\vartheta)$ admit a natural meromorphic continuation from $\Re s \geq 2$ to a larger domain for fixed nonzero ϑ?

Question 17.2 (Functional symmetry)**.** Is there a completion of $\zeta_{\mathrm{H}}(s;\vartheta)$ with a symmetry that reduces at $\vartheta = 0$ to the classical reflection $s \mapsto 1-s$? A nontrivial answer would be required before using the deformation to say anything substantive about the critical line.

Question 17.3 (Spectral zeros)**.** Can the zero set of a continued ζ_{H} be realized as a spectral or resonance set of an operator built intrinsically from $H_{\mu,\sigma,\vartheta}$, rather than as zeros inserted into a multiplier?

Question 17.4 (Classical-zero rigidity)**.** If an analytic two-variable continuation through $\vartheta = 0$ exists, does the exact evenness in ϑ impose any second-order constraint on the motion of the classical Riemann zeros beyond Proposition 12.5?

Question 17.5 (Choice of the second additive coordinate)**.** Which completely additive arithmetic functions $A(n)$, replacing $\Omega(n)$ in $y_n = \kappa A(n)$, produce inequivalent deformations? The choice $A = \Omega$ is distinguished here by its direct relation to the sign of μ on square-free integers, but a classification is open.

Question 17.6 (Higher cohomological curvatures)**.** Determine closed formulas for

$$\frac{d^{2k}}{d\vartheta^{2k}} z_\vartheta(n)\Big|_{\vartheta=0}, \qquad k \geq 2, \tag{17.1}$$

and for their Dirichlet generating series. The fourth-order case is the first test of whether the collapse to finitely many one-prime moments persists or irreducible higher prime geometry appears.

18. Notebook history

Remark 18.1. Version 1 used Möbius-weighted logarithmic translations, Fourier multipliers, an eta multiplier in the critical strip, generalized plane-wave null modes, and a standard CCR non-localization statement. Its main defect was that the arithmetic multiplier and the Heisenberg pair were juxtaposed rather than coupled.

Remark 18.2. Version 2 removes the critical-strip multiplier as the central mechanism and replaces ordinary Dirichlet convolution by the Weyl-cocycle product (6.1). Heisenberg noncommutativity now changes the Möbius inverse coefficients themselves and enters the exact equations of motion of the Möbius Hamiltonian.

Remark 18.3. Version 4 closes the linear-state route structurally and moves the analysis to the cocycle itself. Differentiating the twisted Möbius inverse at $\vartheta = 0$ produces the prime-factor curvature (9.13). Summing it yields (10.3), whose local continuation detects every zeta zero in $\Re s > 1/2$ through (11.2).

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